

2077

Bachelor of Science in Computer Science and Information Technology

Course Title: **Advanced Java Programming**

Code No: **CSC409** Semester: **7th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. What is the significance of stub and skeleton in RMI? Create a RMI application such that a client sends an integer number to the server and the server return the factorial value of that integer. Give a clear specification for every step.
2. You are hired by a reputed software company which is going to design an application for "Movie Rental System". Your responsibility is to design a schema named MRS and create a table named Movie(id, Title, Genre, Language, Length). Write a program to design a GUI form to take input for this table and insert the data into table after clicking the OK button.
3. Describe the responsibility of Serializable interface. Write a program to read an input string from the user and write the vowels of that string in VOWEL.TXT and consonants in CONSOLNANT.TXT.

Section B

Attempt any Eight questions[8×5=40]

4. A non-empty array A of length n is called an array of all possibilities if it contains all numbers between 0 and A_length-1 inclusive. Write a method named isAllPossibilities that accepts an integer array and returns 1 if the array is an array of all possibilities, otherwise it returns 0.
5. Define event delegation model. Why do we need adapter class in event handling?
6. What is the task of Layout manager? Describe about default layout manager.
7. When does the finally block is mandatory in while handling exception? Describe with a suitable scenario.
8. Explain the life cycle of a servlet.
9. What is the task of manifest file? Write the procedure to create it.
10. Why multiple inheritance is not allowed in Java using classes? Give an example.
11. How forms can be created and processed using JSP? Make it clear with your own assumptions.